# Assignment 1: Sampling in your field

CE/EL/MC/CV/GL/GM/ES 361 Probability and Statistics • First Semester, 2026/2027

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**What to hand in, and when**

**One page, handwritten, in your own hand.** Write on one side of a single sheet, with your **name, index number and programme** at the top. Nothing typed is accepted.

There are **two steps, and both are required**:

1. **Online, by 04:00 GMT on Monday 26 October 2026.** Photograph or scan your finished page and upload it here:
   `https://docs.google.com/forms/d/e/1FAIpQLSfGAMoYCY1BYOa9GF4viSF_76DDzs-bgzRGowwJfIl-fK5-XQ/viewform`
   Sign in with your university account. The upload is time-stamped, so late uploads are visibly late. Make sure the writing is legible in the photograph.
2. **On paper, at the first class after the deadline.** Hand in the original sheet. It must be the same page you uploaded.

An upload with no paper, or paper with no upload, is an incomplete submission. Nothing is accepted by email or WhatsApp.

## The task

Explain how sampling is used in **your own programme of study**, and why the choice of sampling technique matters there.

Your page must contain all five of the following.

1. **A named, specific situation from your field** in which somebody has to measure a few things and draw a conclusion about many. Name the place, the process and the quantity measured. "Testing concrete" is not specific; "testing the 28-day strength of concrete poured for the new library foundation at UMaT" is.
2. **Your assigned sampling technique** (see the table below, which depends on your index number). Define it in your own words in two or three sentences, and describe exactly how you would carry it out in the situation from part 1: what the population is, what the sampling frame is, and how a unit is chosen.
3. **Your assigned sample size**, with one sentence on whether you think it is enough for this situation and why.
4. **One honest weakness.** Name a group of units your plan would systematically miss or under-represent, and say which direction that would push the answer: too high, too low, or too certain.
5. **A hand-drawn sketch** of your sampling plan: the site, the population and which units are selected. Rough is fine; it must be drawn by you.

## Your assignment depends on your index number

Take the **last digit** of your index number for the technique, and the **last two digits** for the sample size.

| Last digit of your index number | Your technique |
|---|---|
| 0 or 5 | Simple random sampling |
| 1 or 6 | Systematic sampling |
| 2 or 7 | Stratified sampling |
| 3 or 8 | Cluster sampling |
| 4 or 9 | Convenience sampling |

**Your sample size:** $n = 20 + (\text{last two digits of your index number mod } 30)$

Example: an index number ending in 47 gives $n = 20 + (47 \bmod 30) = 20 + 17 = 37$.

**State your technique and your computed $n$ at the top of the page, with the arithmetic shown.** A page that does not match the writer's index number scores zero, whoever wrote it.

## Marks (20)

| What earns marks | Marks |
|---|---|
| A specific, named situation from your own programme, not a generic one | 5 |
| Correct definition of your assigned technique, correctly applied to that situation | 5 |
| Population, frame and selection rule all clearly stated | 4 |
| An honest weakness, with the direction of the error | 4 |
| A clear hand-drawn sketch | 2 |

Marks are for **specificity**. A page that would read just as well for a student in any other programme earns very few, however fluent it is.

## Honesty: read this part carefully

**This is an individual, handwritten assignment.** You may discuss ideas with classmates and you may read anything you like, but the page you hand in must be written by you, in your own hand, in your own words.

**Artificial intelligence tools must not be used to write any part of it.** You may use them to understand the topic, in the same way you would use a textbook, but text generated by a tool and copied onto your page is not your work and will be treated as such.

**Copying another student's page** means both of you receive zero, the copier and the one who supplied it.

**How this is checked.** Three ways, all simple. First, every page is specific to your index number: the technique and the sample size differ between students, so a shared or generated answer usually fits the wrong brief. Second, the page you hand in on paper is compared with the photograph you uploaded, and the upload is time-stamped. Third, a random selection of students will be asked, in class, to **spend two minutes explaining their own page without looking at it**: where the site is, why that frame, what the weakness was. Students who cannot do that for work bearing their name will have the submission disallowed.

I am not using AI-detection software, because it is unreliable and unfair to students writing in a second language. I am relying on the two checks above, which are not.

**Copy this declaration at the bottom of your page and sign it:**

*"This page is my own handwriting and my own words. I did not copy it from another student or from any source, and I did not use an AI tool to write it."*

Signature _____ Index number _____ Date _____

Sampling is covered in Chapter 1 of the lecture notes, on the course page. Bring questions to the next class rather than guessing: that is what the class is for.